Wind Tunnel Project

Architecture, HMI and control comparison

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8 November 2026

Software demonstrated; physical commissioning pending

Project objectives and operating envelope

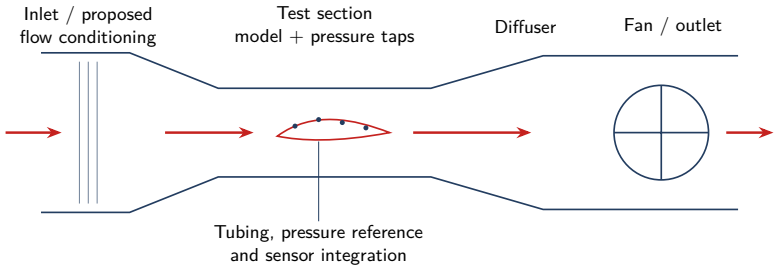
Educational aerodynamic test facility

- Develop the existing open-return tunnel.
- Control airspeed and record repeatable runs.
- Measure $C_p(x/c)$ and pressure-derived lift.
- Compare experiments with XFOIL; explain deviations.

Item	Configured
Airspeed setpoint	0–40 $\mathrm{m/s}$
V20 command	0–50 Hz
Control cycle	100 Hz
Pitot reporting	approximately 20 Hz
Operator interface	Native HMI + local API

Limits are configured values, not demonstrated tunnel performance. V20 system rating: 5.5 kW.

Mechanical integration and airflow quality



Functional concept only: not an as-built drawing and not to scale.

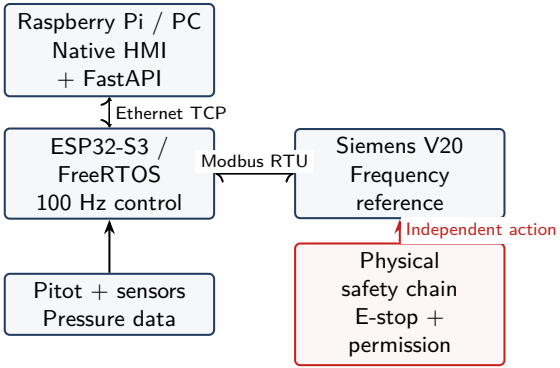
Existing structure

Open-return tunnel; integrate the airfoil, pressure taps, tubing and a stable reference pressure.

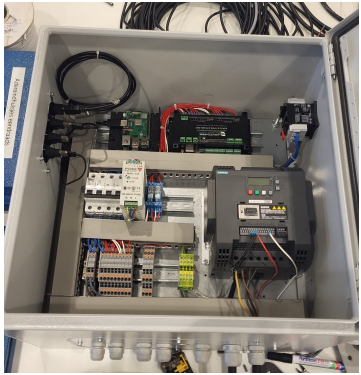
Inspection priorities

Recheck leaks, door closure and flow conditioning identified in the workplan; then measure flow uniformity.

Electrical cabinet and embedded architecture

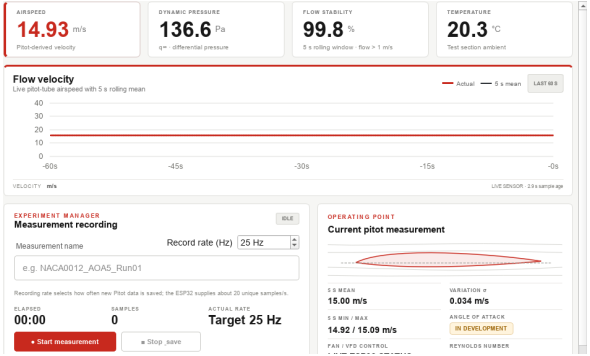


100 Hz control; sensor/control phases separated by 5 ms.
Physical START/STOP/RESET and safety feedback enter the ESP32.



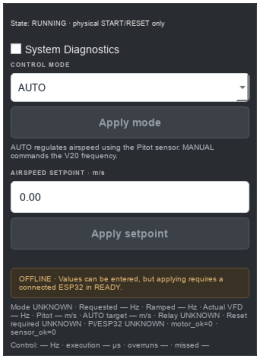
New cabinet – supplied project photo. Assembly progress; commissioning pending.

Current native HMI



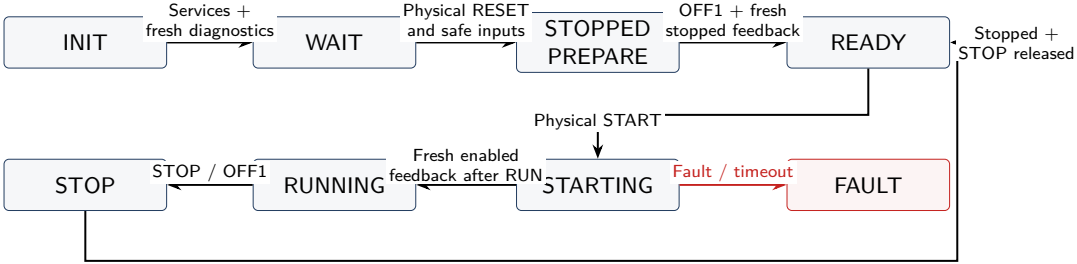
Dashboard: airspeed, pressure, stability, recording and operating point.

Actual widgets with **simulated telemetry**; no hardware connection.



AUTO / MANUAL. Physical START/RESET required.

V20 startup, OFF1/OFF2 and safety



Stopped preparation

- OFF1: `0x047E`; normal ramp stop.
- Intended DO1/In1 path releases OFF2 after RESET and acknowledged OFF1.

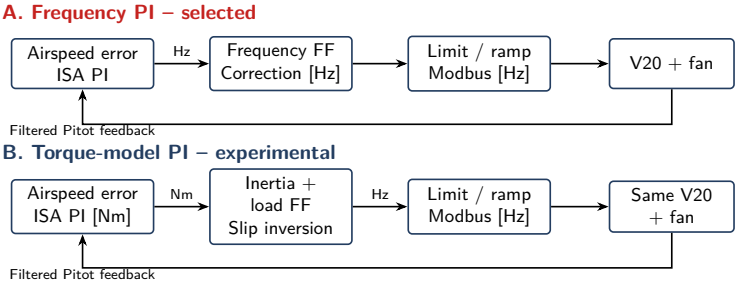
Feedback-confirmed start

- Physical START: `0x047F`, initially zero setpoint.
- RUNNING requires fresh operation-enabled feedback; 5 s start timeout.

OFF3 is kept inactive for normal operation. Applicable faults inhibit RUN; physical safety remains independent.

RESET returns to stopped recovery, never automatic restart.

Two control approaches, one V20 frequency interface

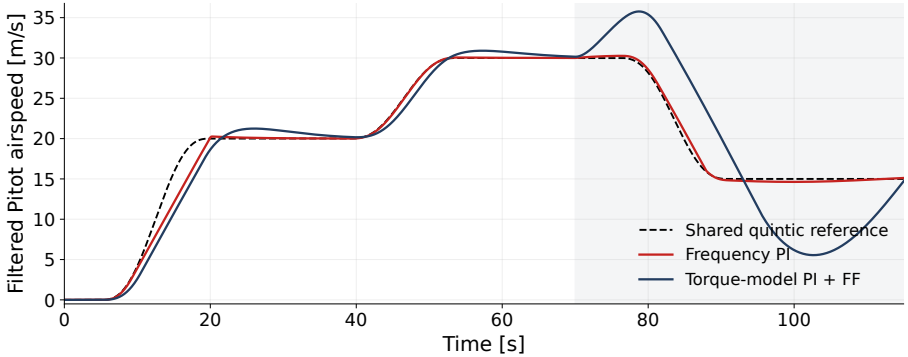


Frequency PI: fewer uncertain physical parameters; identified Hz-to-airspeed model and measured fan characteristic required.

Torque model: inertia/load feedforward and slip inversion; more parameter dependence. Physical torque-model actuation remains disabled.

Both use digital ISA control with Tustin integration. The present comparison uses **PI**, not active derivative action. The V20 receives **Hz**, not torque commands.

Simulated airspeed tracking: shared physical plant

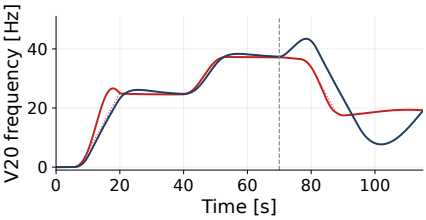


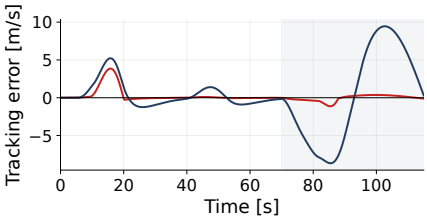
Same initial state; $0 \rightarrow 20 \rightarrow 30 \rightarrow 15$ m/s profiles; 100 Hz; 1 Hz Pitot pressure filter.

Same $J = 0.3712\ \mathrm{kg\,m^2}$, nonlinear motor/fan plant, 0–50 Hz limit and assumed 2 Hz/s drive ramp.

Supplied Python calculation, unchanged defaults. Shading marks a 1 m/s, 0.02 Hz output disturbance from 70 s. Motor/slip/ramp values are unvalidated assumptions.

Control demand, error and interpretation





	Rise [s]	Settle [s]	$MAE_{25-38\,s}$	$MAE_{99-114\,s}$
Frequency PI	9.88	14.66	0.04	0.22
Torque model	9.97	30.59	0.79	7.15

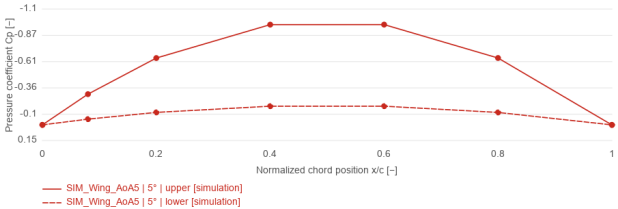
Red: frequency PI; blue: torque-model PI. Dotted: actual Hz. MAE: m/s. Rise: 10–90%; settle: $\pm 2\%$ from $t = 5$ s.

Frequency PI tracks better with these provisional tunings. Feedforward differs; the late window includes a down-transition and disturbance. Hardware validation remains pending.

Aerodynamic acquisition and C_p / C_l analysis

- Pair pressure taps with fresh reference pressure and airspeed.
- Validate channels and integrate pressure over known geometry.
- Compare runs by angle of attack and data source.

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V_\infty^2}$$



Actual analytics widget: synthetic diamond section, $\alpha = 5°$, 14 simulated taps. Not an experimental airfoil result.

Implemented quality gates

Pressure-derived sectional lift/moment only when geometry and data are sufficient. Total drag is not available from pressure alone.

Physical pressure acquisition still needs a configured, validated adapter; real sensor channels and tubing require calibration.

Software validation and project progress

Validation	Passing result
Host regression executables	23 / 23
Firmware build variants	3 / 3
HMI unit/integration tests	104
I/O configuration checks	7
Migration checks	7

Recorded results from software update `b130348`. No physical actuation.

Demonstrated in software

Fresh-start and reset handling; fault lockout; bounded STOP attempts; delayed-reply rejection; stateful drive simulation; HMI acquisition/analysis.

Installation checks still open

DO1/OFF2 wiring and BICO settings; independent safety circuit; V20 watchdog and restart behavior; real sensors, airflow and dynamics.

Next steps and conclusions

1. Verify cabinet wiring, independent safety and V20 OFF2/watchdog behavior with the motor isolated.
2. Identify actual frequency dynamics and validate the stored fan calibration.
3. Commission controlled starts/stops, sensor channels and airflow quality.
4. Acquire repeatable C_p data and compare with XFOIL and uncertainty bounds.

Current maturity

Software and HMI extensively tested; new cabinet assembled.
Physical commissioning and experimental aerodynamic validation remain pending.